

Replication of Computerized Assessment of Motor Imitation (CAMI) with neurodivergent pediatric population

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Introduction

- **Motion imitation is crucial to social bond formation and learning.**
- Imitation deficits are prevalent in **ASD** and have been linked to **social communication impairments**.
- Human observation coding (HOC) in imitation evaluation is **time consuming and error-prone**

ORIGINAL STUDY (Summary)



- Tunçgenç et al proposed an objective, continuous and scalable score of Computerized Assessment of Motor Imitation (CAMI) with a range of 0-1
- They found that CAMI (accuracy = 87.2%) outperformed HOC (accuracy = 74.4%) in distinguishing children with ASD and typically developing children.

RESEARCH QUESTION

On replication of CAMI with neurodivergent pediatric population, is there a difference in CAMI score for children with ADHD in presence of heterotypic comorbidities?

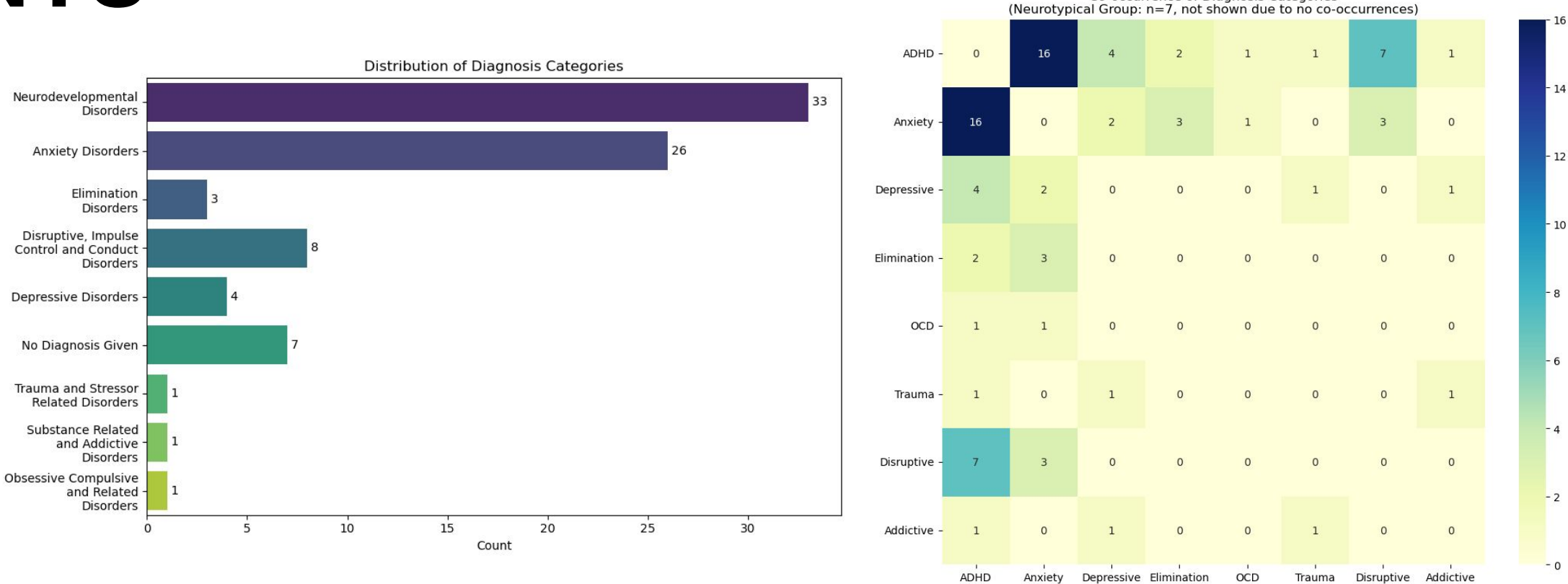
HYPOTHESIS

There will be no significant difference in CAMI score for children with ADHD and heterotypic comorbidity compared to neurotypical children.

Methods

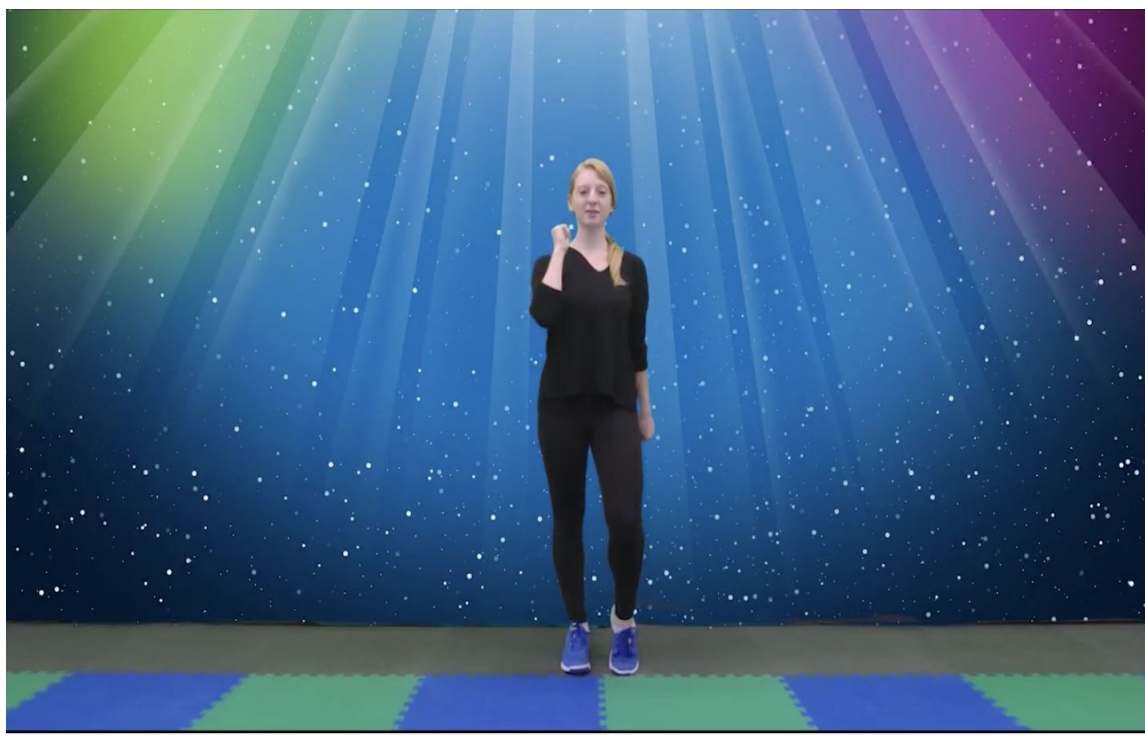
PARTICIPANTS

- N = 50
- 23F, 27M
- Age = 10.37 yrs (SD=2.40)



PROTOCOL

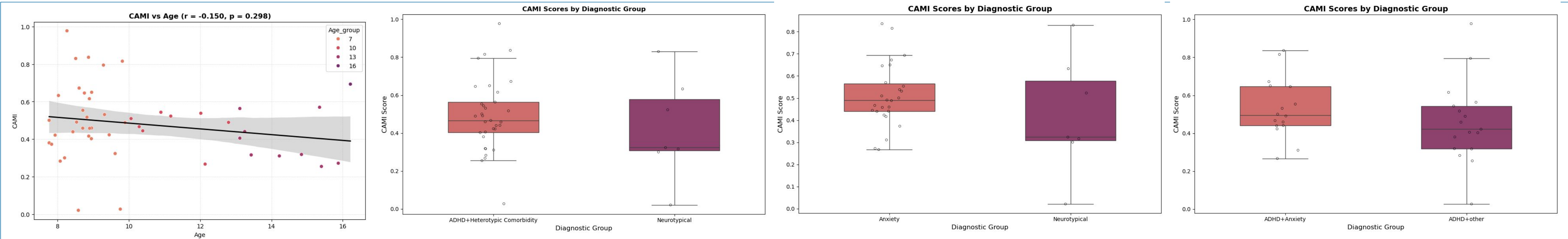
- Microsoft Kinect Camera - Front, Rear
- Assessments - SAS, SCQ, SRS, RBS, C3SR, SWAN, WHODAS-P, WHODAS-SR
- Two dance sequences -
 - Sequence 1 at 100% speed (PT_100)
 - Sequence 2 at 100% speed (PT1_100)
 - Sequence 1 at 60% speed (TR_60)



DATA PROCESSING



Results



KEY OUTCOMES

- No correlation between CAMI score and Age.
- No correlation between CAMI scores and Cognitive assessment scores.
- Similar imitation performance in children with ADHD and heterotypic comorbidity, and neurotypical children
- Similar imitation performance in children with Anxiety and heterotypic comorbidity, and neurotypical children
- Similar imitation performance in children with ADHD and heterotypic comorbidity (except for Anxiety) and ADHD and heterotypic comorbidity (with Anxiety)

ALIGNMENT WITH PRIOR EVALUATION STUDY !

Santra et al. found that children with ASD showed poorer CAMI performance than neurotypical children, regardless of co-occurring ADHD. Also, they found that children with only ADHD showed similar CAMI performance to neurotypical children. Above results are consistent with those of Santra et al for children with ADHD.

Study Takeaway

Motion imitation deficits measured using CAMI are not observed in children with ADHD regardless of heterotypic comorbidities.



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References:
[1] Tunçgenç, 2022
[2] Santra, 2026
[3] Alexander, 2017